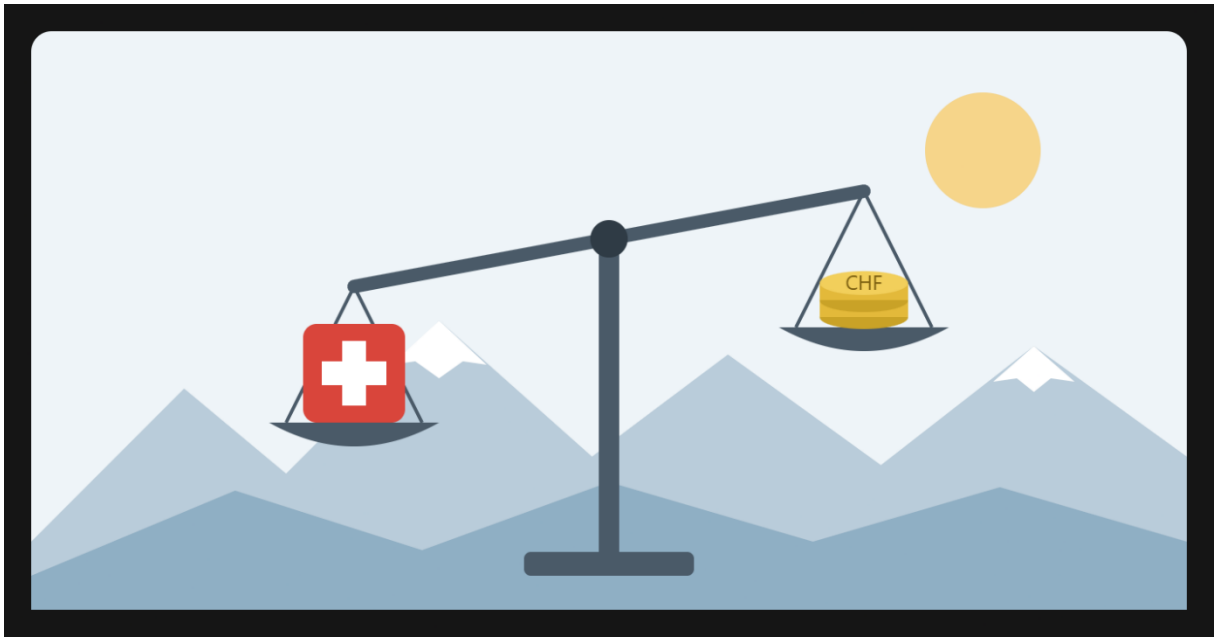
HSLU Hochschule Luzern

Premiums vs. Paychecks: Health Insurance Affordability Across Swiss Cantons, 2012–2022



Balance between health insurance premiums and income. Created with Claude (Anthropic, 2026)

Project Documentation

Data Collection, Integration and Preprocessing

Autum semester 2026/2027

Alisa Bitschnau, Chaitanya Arya, Jobin Joseph

1. Introduction

Mandatory health insurance in Switzerland is financed by premiums that are charged per insured person. Insurers set them by canton and premium region, but not by income (Art. 61 KVG). The same premium therefore weighs very differently on households, especially because both premium levels and incomes differ strongly between cantons. Rising premiums are one of the main public concerns in Switzerland. In the UBS Worry Barometer 2025, 45 % of respondents named health care and health insurance premiums as one of the country's most important problems, more than any other issue (Schäfer, 2026). In June 2024, voters rejected a popular initiative that would have limited premiums to 10 % of disposable income (FOPH, 2024). This project therefore combines premium data with cantonal tax statistics to measure how heavy the premium is relative to income between 2012 and 2022, the latest year with published tax statistics.

To measure this burden, we ask three research questions.
RQ1 - Change over time: How has the affordability of mandatory health insurance changed between 2012 and 2022?

RQ1.1 Did premiums increase faster than income?

RQ2 - Regional differences: How does affordability differ between the cantons, and have these differences grown or narrowed?

RQ3 - Insurance choices: How much do premiums differ by deductible and insurance model across the cantons?

2. Methods

2.1. Data sources and acquisition

Our project is based on four public federal sources (Table 1). The income data come from the tax statistics of the Federal Tax Administration (ESTV) and from the gross wages of the Federal Statistical Office (BFS). The premium data come from the yearly premium overviews and the premium calculator of the Federal Office of Public Health (FOPH) on priminfo.admin.ch. All data were collected with Python. Before scraping a website, robots.txt was checked. The ESTV allows access, and priminfo.admin.ch has no robots.txt, so nothing is disallowed. Our scrapers pause between requests, log every download and store all raw files unchanged in data/raw/.

Source	Method	Data
ESTV, direct federal tax statistics	Static scraping (requests, BeautifulSoup): XLSX links read from the overview page	Taxable income and taxpayers per canton and household type, 2012–2022 (11 files)
BFS STAT-TAB (PxWeb)	REST API, JSON POST queries	Median, 25th and 75th percentile of monthly gross wages by major region (2012–2022; biennial)
FOPH priminfo archive	Static scraping of the archive page, PDF download, parsing with pdfplumber	Official premium tables of all insurers, regions, age groups, models and deductibles, 2012–2022 (11 PDFs)
FOPH priminfo premium calculator	Dynamic scraping with Selenium	Current premium of every insurer in the capital of each canton (2026, 628 offers)

Table 1: Data sources, acquisition methods and responsibilities

The income data come from the ESTV, which publishes one Excel file per tax year with key figures for every canton. Because the download links contain random identifiers, the scraper reads the overview page with BeautifulSoup and selects the files for 2012–2022 by their name. In addition, we queried the BFS STAT-TAB API with JSON POST requests for the median monthly gross wage per large region.

The premiums for 2012–2022 come from the yearly premium overviews on the FOPH archive page. We collected the links with BeautifulSoup and downloaded the 11 PDFs, each with more than 1,100 pages. The tables have no cell borders, so pdfplumber could not detect them. We therefore read each page word by word with its position. Model, canton, age group and region come from the page header, and each premium is assigned to its deductible by its position under the column header, because some insurers leave deductibles out. The layout also differs between cantons and years, for example Italian headers in Ticino. In total we extracted 2,922,813 premium values. Only 16 lines (2016–2018) could not be assigned safely and were left out.

The premium calculator is a JavaScript-based interactive form. We used Selenium to automate the data entry, including selecting the postal code from the suggestion list, entering the birth year, choosing the deductible and specifying accident cover and the insurance model. Selenium then submitted the form and extracted the premiums from the results table. The data were collected in October 2026 for a standard adult profile (born in 1988, CHF 300 deductible, accident cover included, standard model) in all 26 cantonal capitals. As they represent only one premium region per canton and the years 2023–2025 are missing, they are treated as a snapshot of current premiums and are not part of the figures.

2.2. Cleaning and transformation

Table 2 shows how we applied the six required cleaning steps to both datasets. Every script prints the result of each check, so the cleaning can be reproduced. The 3xIQR rule flags values far outside the middle half of the data. Flagged values were checked but not deleted.

Step	Income (ESTV, BFS)	Premiums (priminfo PDFs)
Gaps / missing data	All 286 canton-years present, no missing values	No missing values; all 26 cantons in every year. HMO not offered in GL and VS before 2016.
Data types	Cast explicitly; changing column names mapped to one schema	Year, region, deductible as integers, premiums as floats, model and canton as categories
Value ranges	Positive values, P25 ≤ median ≤ P75	All premiums > 0. Premiums fall with higher deductibles and are lower without accident cover; only exception (BeneFit, CHF 1,500) was checked in the PDF and kept.
Outliers	3xIQR on yearly change: Valais 2020 flagged and kept	3xIQR within year and deductible: 1,788 high values (mostly GE, BS) are real tariffs and kept
Format / merge	Yearly files stacked; household types in long format	One value per canton and year for the reference case (see 2.3); merged with income on canton and year (286 rows).
Enrichment	Large region, growth index (2012 = 100), BFS wage	Annual premium, premium burden, discount vs. reference case, yearly saving

Table 2: Cleaning and transformation steps

2.3. Analysis methods

To ensure consistent and comparable results, we define a standard reference case, an adult in the standard model with the lowest deductible (CHF 300, the yearly amount people pay themselves before the insurance pays) and accident cover. For each canton and year, we first take the mean premium over all insurers within a premium region and then the mean over the regions of the canton, so that regions with many insurers do not count more. Using the median instead of the mean changed the premiums by less than 1 % on average and left the ranking of the cantons practically unchanged (rank correlation of at least 0.99 in every year).

Our primary indicator is the premium burden, the annual premium divided by the median taxable income, multiplied by 100. We use the median rather than the mean because income distributions are right-skewed. Since the calculation compares the premium of one adult with the income of one tax unit (a married couple counts as one unit), the burden is not the actual share of household income spent on health insurance but a comparative indicator across cantons and years. To compare how fast premiums and incomes grew, we set both to 100 in 2012, so a value of 130 means an increase of 30 %. The line shows the typical (median) canton, and the shaded band the range in which most cantons lie (10th to 90th percentile). Regional differences are measured by the range and standard deviation of the burden and by the Pearson correlation between income and premium. For RQ3 we apply the same averaging to all four insurance models and six deductibles and compute the discount against the reference case. The analysis uses pandas and NumPy, and the figures are made with matplotlib and Plotly. All code, data and two interactive dashboards, including a map of the cantons and a heatmap, are available in our GitHub repository (Arya et al., 2026).

3. Results and discussion

3.1. RQ1: Premiums grew almost three times faster than income

Between 2012 and 2022 the premium of the reference case rose by +33.8 % in the median canton (from CHF 4,151 to CHF 5,459 a year), while median taxable income rose by only +12.3 % (Figure 1).

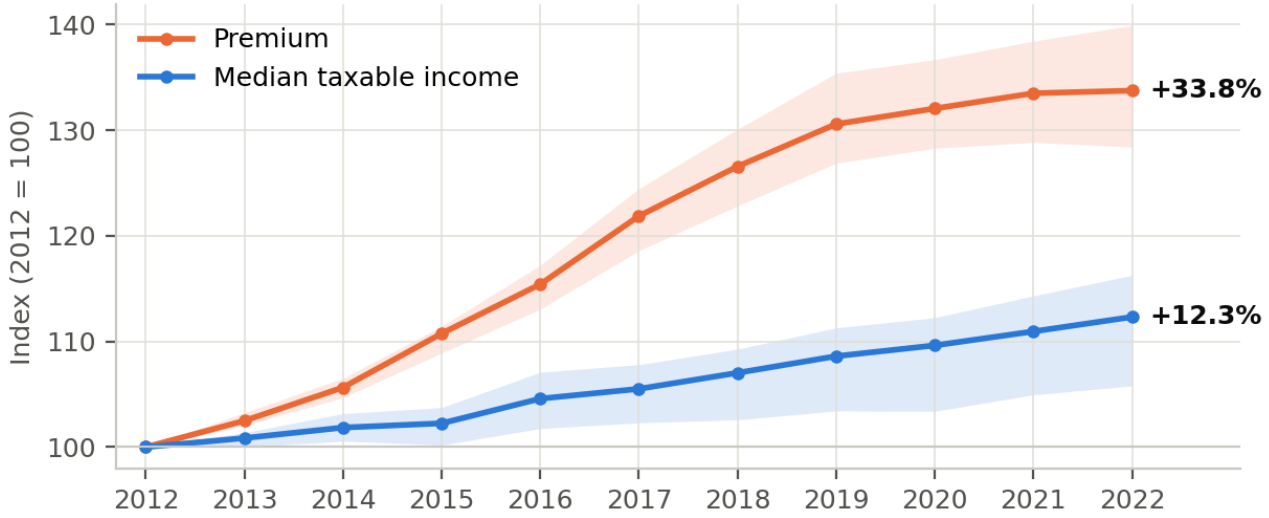


Figure 1: Premium and median taxable income as an index (2012 = 100). Lines: median canton; bands: 10th–90th percentile of the 26 cantons

Premiums grew faster than income in all 26 cantons, from +22 % in Bern to +44 % in Appenzell Ausserrhoden. From 2013 on, the two shaded bands no longer overlap. This means that even in the cantons where premiums grew slowly, they grew faster than income in the cantons where income grew quickly. As a result, the typical burden rose from 9.3 % to 11.3 % of median taxable income, an increase of two percentage points. The rise was steepest between 2015 and 2019. In 2021 and 2022 the burden eased slightly (11.5 % → 11.3 %) because premiums hardly changed while incomes kept growing.

3.2. RQ2: The gap between cantons has widened

The burden differs strongly between cantons (Figure 2). In 2022 it ranges from 7.6 % in Zug to 17.5 % in Geneva. It is highest in western Switzerland (Geneva, Neuchâtel, Jura, Vaud), Ticino and Basel-Stadt, and lowest in central Switzerland. Every canton has a higher burden in 2022 than in 2012, but the increase is uneven. It rose most in Geneva (+4.6 percentage points, pp) and Neuchâtel (+4.2 pp), and least in Zug (+0.8 pp) and Bern (+0.9 pp). In Geneva the median taxable income was even 1.3 % lower in 2022 than in 2012. This does not mean that incomes fell. The ESTV statistic does not include people who are taxed at source (mainly foreign workers whose tax is deducted directly from their salary), and in Geneva the number of counted taxpayers grew by 26.5 %, far more than in other cantons. Most of the new taxpayers have low incomes, which pulls the median down, while the mean still rose by about 5 % and the regional BFS wage by 4.4 %. Geneva's burden increase is therefore likely overstated. Overall, the differences between cantons have widened: the range grew from 6.8 to 9.9 percentage points and the standard deviation from 1.9 to 2.7 pp. Even without Geneva, the gap widened (range 6.8 → 8.8 pp, standard deviation 1.9 → 2.4 pp), so this result does not depend on the Geneva data.

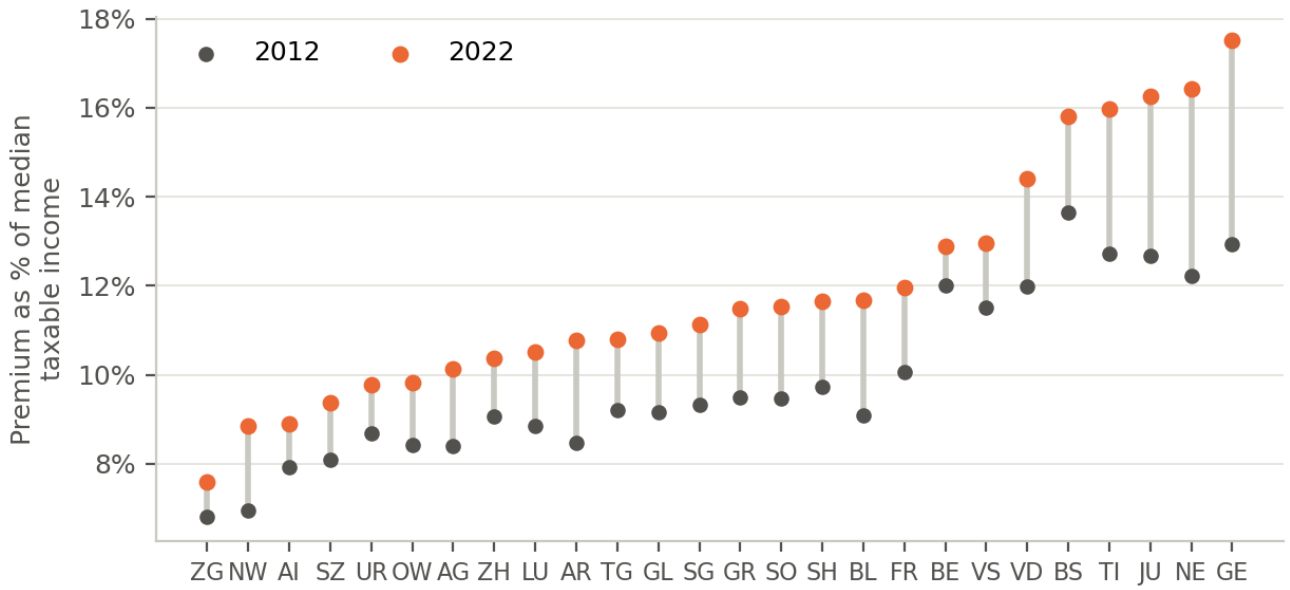


Figure 2: Premium burden per canton in 2012 and 2022, sorted by the burden in 2022

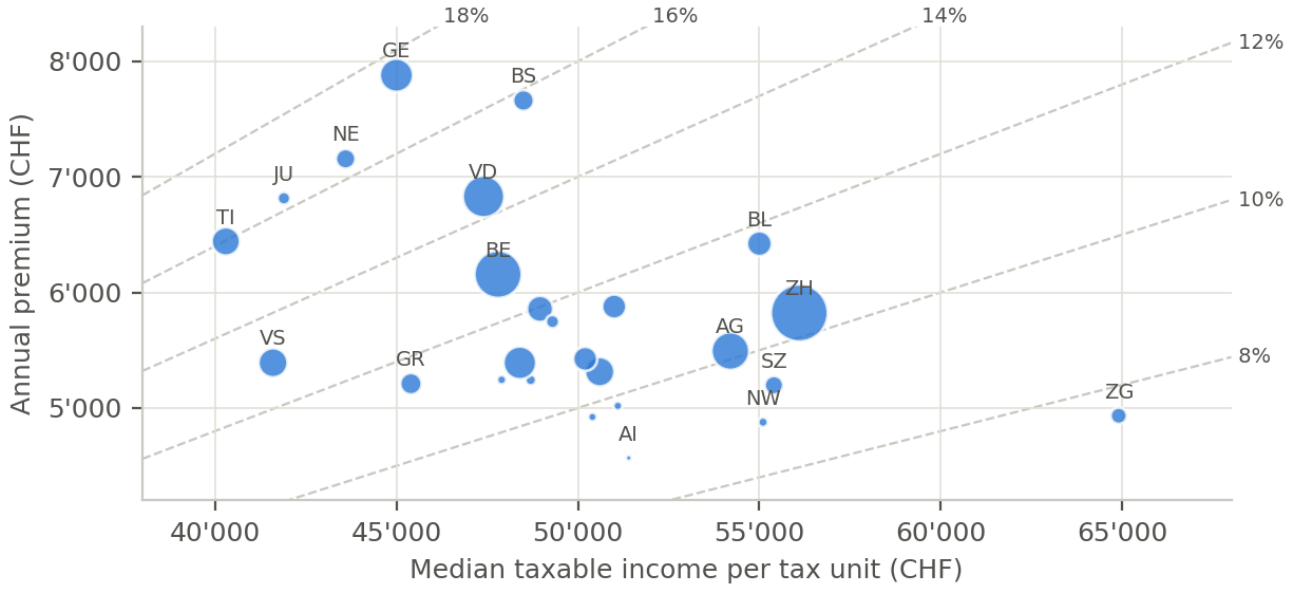


Figure 3: Median taxable income and annual premium per canton in 2022. Bubble size: number of taxpayers. Dashed lines: equal burden

Part of this pattern is explained by the link between income and premiums. Cantons with lower median incomes tend to have higher premiums (Figure 3), so both effects increase the burden instead of balancing each other. This relationship has become stronger (r ranges from −1 to +1; negative values mean that higher incomes go with lower premiums). The correlation between median income and premium was close to zero in 2012 (r = −0.10) and clearly negative in 2022 (r = −0.46). Geneva, Neuchâtel and Jura combine low median incomes with high premiums, while Zug combines the highest income with low premiums.

3.3. RQ3: Insurance choices help, but not where it matters most

In 2022, all alternative insurance models were about 11–12 % cheaper than the standard model at the same deductible (Figure 4a). In 2012, the HMO model was still 17 % cheaper. A higher deductible saves even more. In the typical canton, choosing the CHF 2,500 deductible instead of CHF 300 lowers the annual premium of the standard model by CHF 1,535. This saving is almost the same in every canton and year (CHF 1,434–1,560), because the law allows a discount of at most 70 % of the extra costs the insured person takes on, which is CHF 1,540 (Art. 95 KVV; FOPH, n.d.). Since premiums kept rising while this amount stayed the same, the discount fell from 37 % of the premium in 2012 to 28 % in 2022.

The higher deductible only pays off if the year's health costs stay low. With the CHF 2,500 deductible, insured people pay a larger part of their costs themselves. This option is therefore only cheaper if the yearly health costs stay below about CHF 2,000 a year. Combining the cheapest model with the highest deductible lowers the typical burden from 11.3 % to 6.8 % (Figure 4b). However, even with this cheapest choice, the burden in 8 of 26 cantons (Geneva, Neuchâtel, Basel-Stadt, Jura, Ticino, Vaud, Bern, Valais) remains higher than the standard burden in Zug. Where someone lives matters more for the burden than how they insure, and the cheapest options are mainly useful for people who rarely need care.

Dropping accident cover lowers premiums by about 6 % but leaves the ranking of the cantons by burden practically unchanged (Spearman rank correlation at least 0.998 in every year), so our conclusions do not depend on this choice.

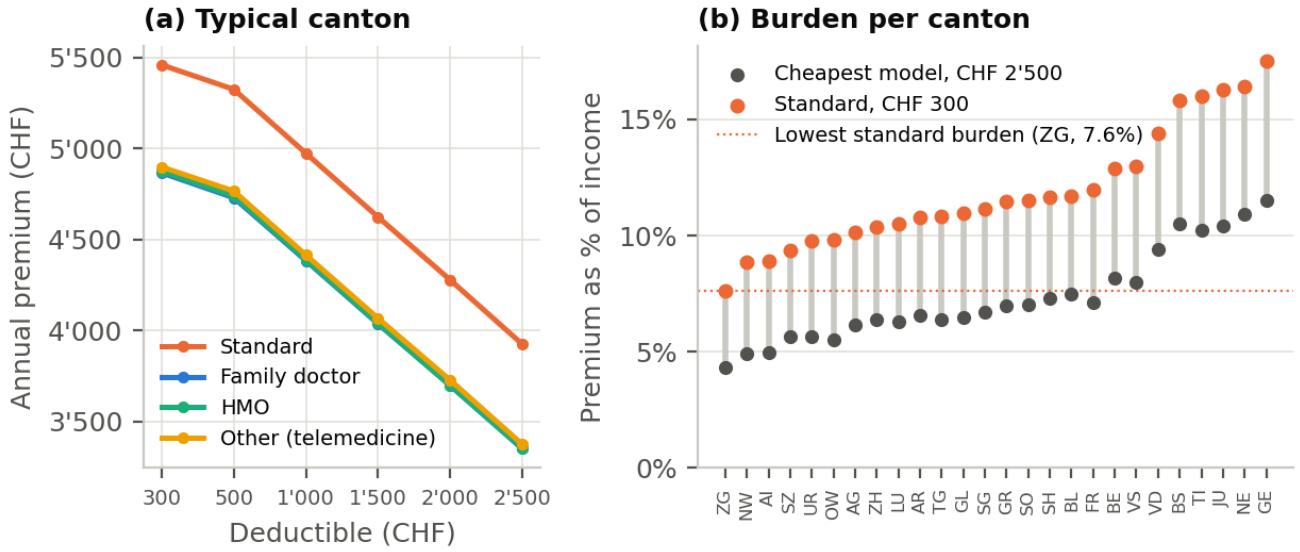


Figure 4: (a) Annual premium by insurance model and deductible, median canton 2022. (b) Burden with the reference case and with the cheapest model at CHF 2,500 deductible, per canton 2022

4. Conclusion

Between 2012 and 2022, health insurance premiums grew almost three times as fast as median taxable income in Switzerland, and the typical burden rose from 9.3 % to 11.3 %. The burden grew in every canton, but most in western Switzerland and Ticino, where incomes are lower and premiums higher, so the gap between cantons widened. Alternative insurance models and high deductibles can lower the premium by about 39 % in the typical canton, but the deductible discount is fixed in francs and loses weight as premiums rise, and it shifts costs to people who get sick.

Most of the work went into data acquisition and cleaning rather than analysis. The PDF layouts changed from year to year, so the parser had to read word positions instead of tables, and the tax files had to be mapped to one common structure. Systematic range and outlier checks paid off, because they revealed a real tariff quirk and a likely break in the Valais tax data.

Taxable income is income after deductions and per tax unit, not disposable household income. Premium subsidies, which reduce the burden for low incomes, are not included. Insurers and premium regions are weighted equally because market shares and population per region are not available. The 2020 value of Valais is flagged as a likely break in the source data. The calculator snapshot covers only the capital of each canton. In Geneva, more low-income taxpayers lower the median, so its burden increase is likely overstated (see 3.2).

The analysis could be extended with premium subsidies per canton (priminfo publishes them as PDFs), with population-weighted premium regions, with the ESTV data for 2023 and later once published, and by running the Selenium scraper for all premium regions and household types. A regression could then explain the burden with income, urbanisation and healthcare supply per canton.

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